



# Kalpana Thirumenirasa

Software Developer

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Motivated Software Developer with an MSc in Applied Computer Science and hands on experience in building automation pipelines, data processing systems, and cybersecurity research tools. Strong expertise in Python for scripting, backend development, and data analysis. Experienced in Docker, CI/CD, and working with APIs and databases. Passionate about writing clean, efficient code, solving complex technical problems and contributing effectively within Agile teams.

## Education

**MSc Applied Computer Science**, Deggendorf Institute of Technology, Germany Jan 2026

GPA: 2.7

Master's Thesis: Comparing LLM Based Fuzzing Techniques for Software System Robustness

- Conducted a comparative study of fuzzing frameworks on deep learning libraries.
- Designed automated pipelines using Python, Docker, and Conda.
- Evaluated system robustness using crash analysis and code coverage.

**Key Modules:** Software Engineering, Embedded Systems & Security, Database Systems, Computer Networks, Probability & Statistics

**BSc (Hons) Computer Science and Technology**, Uva Wellassa University of Sri Lanka Jan 2022

GPA: 2.17

Bachelor Thesis: Sentiment Analysis on Crowd Behaviour

- Built a machine learning model using TF IDF and Support Vector Machines (SVM) achieving 84% accuracy on Twitter datasets.
- Performed data preprocessing, feature extraction, and model evaluation using Python.

**Key Modules:** Data Structures and Algorithms, Database Systems, Software Engineering, Operating Systems, Computer Networks, Artificial Intelligence, Machine Learning, Web Development, Object Oriented Programming

**Diploma in Information Technology**, ESOF Metro College Jan 2017

## Professional Experience

**Working Student – Deggendorf Institute of Technology** Germany 2024 – 2025

- Contributed to cybersecurity and protocol fuzzing research projects.
- Improved Python based scripts and maintained version controlled repositories.
- Monitored network traffic using Wireshark for analysis and debugging.

**Management Trainee (SAP ABAP) – Michelin Business Services Lanka (Pvt) Ltd** Sri Lanka Jan 2023 – Sep 2023

- Developed and supported SAP ABAP programs to enhance business workflows.
- Assisted in debugging and optimizing ABAP code.
- Collaborated with cross functional teams to translate business needs into SAP solutions.
- Built workflow automation solutions integrated with SharePoint.
- Worked in Agile/Scrum environments for efficient delivery.

**Software Engineer Intern – Creative Software**

Sri Lanka Apr 2022 – Sep 2022

- Developed and tested applications ensuring reliability.
- Built a .NET based web application using C#, Docker, and PostgreSQL.
- Participated in Agile development processes and code reviews.

## Skills

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**SAP Technologies:** SAP ABAP, Debugging, Reports, Enhancements

**Programming Languages:** ABAP, Python, Java, C, C#, C++

**Tools:** SAP GUI, Git, Jira, Docker, CI/CD

**Databases:** MySQL, PostgreSQL, MongoDB

**Concepts:** Agile/Scrum, OOP, Clean Code, REST APIs

**Operating Systems:** Linux, Windows

## Projects

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### SAP ABAP Mini Project

Developed basic ABAP reports using Open SQL, internal tables, and selection screens.

### City Guide Application

Full-stack web app using React, Next.js, and MongoDB.

### Traffic Sign Detection

YOLO-based object detection project with model tuning.

## Languages

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Tamil (Native) | English (C2) | German (B1)

## Personal Interests

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- Problem solving and logical puzzles
- Learning new technologies
- Listening to music
- Cultural activities and social engagement
- Traveling and exploring new places

Passau, Germany



**Kalpana Thirumenirasa**